Gaurav Anand

gauravanand2506@gmail.com | +91-7717452178 | linkedin.com/in/gaurav-anand | github.com/gauravanand10

Education

Thapar Institute of Engineering & Technology, Patiala *Aug 2023 – Present*
Bachelor of Technology in VLSI Design; CGPA: 8.17/10 *Patiala, India*
Relevant Coursework: Digital System Design, Embedded Systems, Computer Architecture, CMOS VLSI Design

Technical Skills

Languages: Verilog, SystemVerilog, TCL, Python, MATLAB
EDA & Verification Tools: Synopsys (VCS, Verdi, Design Compiler, SpyGlass); Cadence (Genus, Conformal LEC, Modus, Xcelium); Xilinx Vivado
Protocols: AMBA-AHB, AXI, UART, JTAG
Architecture & Simulation: Gem5, RISC-V GNU Toolchain, Spike

Projects

RV32I Processor Performance Modelling *Jul 2026 – Present*
Gem5, RISC-V GNU Toolchain, Spike, Verilog

- Developing a 5-stage RV32I processor and modelling its microarchitecture using Gem5.
- Analyzing IPC, CPI, execution cycles, and memory behaviour across RISC-V workloads.
- Exploring cache hierarchy and memory-system configurations for performance optimization and future RTL/FPGA implementation.

Systolic Array Accelerator for Real-Time Sobel Edge Detection *Jun 2026 – Jul 2026*
VCS, Verdi, Genus, Conformal LEC, Modus, Spike, Vivado, MATLAB [GitHub]

- Designed a streaming systolic-array Sobel accelerator in Verilog with parallel G_x/G_y computation, verified using VCS and Verdi.
- Synthesized using Cadence Genus at 180nm, achieving 32,376 cell area, 44,328 total area, and $\sim$7.02mW power.
- Performed RTL-to-gate Conformal LEC and scan-cell insertion using Cadence Modus.
- Evaluated RISC-V workload execution using Spike and deployed on Zybo-7000 for a 640$\times$480 grayscale image, reconstructed in MATLAB.

Low-Power Banked Memory Architecture *Apr 2026 – May 2026*
Verilog, VCS, Verdi, Genus, Design Compiler [GitHub]

- Designed a banked memory architecture with a bank controller and address decoder to reduce dynamic power by activating only the required bank.
- Implemented in Verilog with configurable number of banks and tested using VCS/Verdi for functional verification across random access patterns.
- Synthesized using Genus at 180nm, analyzing area and power, and achieved significant reduction in switching activity compared to a monolithic memory.

AMBA-AHB Protocol Controller Design & Verification *Feb 2026 – Mar 2026*
VCS, Verdi, SpyGlass, Design Compiler [GitHub]

- Designed an AMBA-AHB bus architecture with Master, Slave, Decoder, and Multiplexer modules supporting single, INCR4, and WRAP4 burst transactions, verified using VCS and Verdi.
- Performed SpyGlass RTL linting and analyzed QoR, timing, area, power, and hierarchy reports.
- Achieved 100MHz timing closure with zero setup/hold violations.
- Integrated 4$\times$256$\times$32 SRAM macros, reducing standard cell count from 95K to $\sim$7K and power from 172mW to 30–40mW.

Internship

Patiala Locomotive Works *Jun 2025 – Jul 2025*
Vocational Trainee *Patiala, Punjab*

- Learned and executed the procedure to test integrated circuits using the Advance Diagnostic Tester, ensuring accurate functionality verification.

Achievements

50-Day RTL Design Challenge *Jun 2026 – Present*
[GitHub]

- Completing a 50-day RTL design challenge by designing and verifying one Verilog problem daily.